# GAUSS-MANIN CONNECTION FOR FAMILY OF ELLIPTIC CURVES

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## 1. de Rham cohomology and Gauss-Manin connection

Let $X$ a smooth separated scheme of finite type over a Noetherian ring $Spec(R)$, then the sheaf of Kahler differentials is a locally free quasi-coherent module $\Omega^1_{X/R}$. The derivation $d$ defines the complex

$$\Omega^\bullet_{X/R} : 0 \to \mathcal{O}_X \to \Omega^1_{X/R} \to \Omega^2_{X/R} \to \cdots$$

and the cohomology of the complex obtained applying the functor $\Gamma = \Gamma(X, -)$ takes the name of *de Rham cohomology* $H_{dR}(X/R)$. Consider now the relative case $\pi : X \to S$ smooth morphism of varieties over a field $k$. Then we can consider the sheaf of relative differentials $\Omega^\bullet_{X/S} = \Omega^\bullet_{X/k}/\pi^*\Omega^\bullet_{S/k}$ and define the *relative de Rham cohomology* $\mathcal{H}^\bullet_{dR}(X/S)$ as the right derived hypercomology

$$\mathcal{H}^q_{dR}(X/S) = \mathbb{R}^q\, \pi_*(\Omega^\bullet_{X/S}).$$

From the short exact sequence defining the relative differentials

$$0 \to \pi^*\Omega^1_{S/k} \to \Omega^1_{X/k} \to \Omega^1_{X/S} \to 0$$

and using the fact that the sheaves are locally free, we obtain a filtration

$$\Omega^\bullet_{X/k} = F^0\Omega^\bullet_{X/k} \supseteq F^1\Omega^\bullet_{X/k} \supseteq F^2\Omega^\bullet_{X/k} \supseteq \cdots$$

with

$$F^i\Omega^\bullet_{X/k}/F^{i+1}\Omega^\bullet_{X/k} = \pi^*\Omega^i_{S/k} \otimes_{\mathcal{O}_X} \Omega^{\bullet-i}_{X/S}$$

The fact that the sheaves are locally free implies that there is a trivializing cover of affine schemes $\{U_i\}_i$ where

$$\Omega^1_{X/k|U_i} \cong \pi^*\Omega^1_{S/k|U_i} \oplus \Omega^1_{X/S|U_i}$$

and then taking exterior powers we obtain

$$\Omega^n_{X/k|U_i} \cong \bigoplus_{i=0}^{n} \left(\pi^*\Omega^i_{S/k|U_i} \otimes \Omega^{n-i}_{X/S|U_i}\right).$$

The filtration is then locally given by

$$F^j\Omega^n_{X/k|U_i} = \bigoplus_{i=0}^{n-j} \left(\pi^*\Omega^i_{S/k|U_i} \otimes \Omega^{n-i}_{X/S|U_i}\right).$$

Since the relative de Rham sheaf is defined to be the sheaf associated to the presheaf that locally assigns

$$U \mapsto \mathbb{H}^q(\pi^{-1}(U), \Omega_{X/S|\pi^{-1}(U)})$$

we can then assume $S = Spec(R)$ to be affine and define the Gauss-Manin connection locally.

Considering the spectral sequence $E_r^{p,q}$ associated to the filtration we have that the first page is given by

$$
\begin{array}{cccc}
\Omega^2_{X/S} & \pi^*\Omega^1_{S/k}\otimes_{\mathcal{O}_X}\Omega^2_{X/S} & \pi^*\Omega^2_{S/k}\otimes_{\mathcal{O}_X}\Omega^2_{X/S} & \pi^*\Omega^3_{S/k}\otimes_{\mathcal{O}_X}\Omega^2_{X/S} \\
\uparrow & \uparrow & \uparrow & \uparrow \\
\Omega^1_{X/S} & \pi^*\Omega^1_{S/k}\otimes_{\mathcal{O}_X}\Omega^1_{X/S} & \pi^*\Omega^2_{S/k}\otimes_{\mathcal{O}_X}\Omega^1_{X/S} & \pi^*\Omega^3_{S/k}\otimes_{\mathcal{O}_X}\Omega^1_{X/S} \\
\uparrow & \uparrow & \uparrow & \uparrow \\
\mathcal{O}_X & \pi^*\Omega^1_{S/k} & \pi^*\Omega^2_{S/k} & \pi^*\Omega^3_{S/k} \\
\uparrow & \uparrow & \uparrow & \uparrow \\
0 & 0 & 0 & 0
\end{array}
$$

The second page is given by

$$0 \longrightarrow H^2_{dR}(X/R) \longrightarrow \Omega^1_{R/k}\otimes_R H^2_{dR}(X/R) \longrightarrow \Omega^2_{R/k}\otimes_R H^2_{dR}(X/R) \longrightarrow \Omega^3_{R/k}\otimes_R H^2_{dR}(X/R)$$

$$0 \longrightarrow H^1_{dR}(X/R) \longrightarrow \Omega^1_{R/k}\otimes_R H^1_{dR}(X/R) \longrightarrow \Omega^2_{R/k}\otimes_R H^1_{dR}(X/R) \longrightarrow \Omega^3_{R/k}\otimes_R H^1_{dR}(X/R)$$

$$0 \longrightarrow H^0_{dR}(X/R) \longrightarrow \Omega^1_{R/k}\otimes_R H^0_{dR}(X/R) \longrightarrow \Omega^2_{R/k}\otimes_R H^0_{dR}(X/R) \longrightarrow \Omega^3_{R/k}\otimes_R H^0_{dR}(X/R)$$

## 2. Gauss-Manin connection for family of Elliptic Curves

2.1. **A CM family.** Consider the family of elliptic curves $\mathcal{E} : y^2 = x^3 - t$ over $S = Spec(\mathbb{C}[t, t^{-1}])$. Now recall that a basis for the de Rham cohomology of an elliptic curve is given by

$$H^1_{dR}(\mathcal{E}/S) = \langle \omega, \eta \rangle$$

In particular we have

$$\omega = \frac{dx}{y}, \quad \eta = \frac{xdx}{y}.$$

Given the equation $y^2 = x^3 - t$ we have that $\Omega^1_{\mathcal{E}/S}$ is a free $R$-module where

$$2ydy - 3x^2dx = 0.$$

Taking $\omega = \frac{2x}{3t}dy - \frac{y}{t}dx$ we can observe that

$$\begin{cases} dx = y\omega, \\ dy = \frac{3x^2}{2}\omega \end{cases}$$

but in the full module $\Omega^1_{\mathcal{E}/\mathbb{C}}$ we have that the relation lifts to

$$2ydy - 3x^2dx + dt = 0.$$

In particular the previous relations become

$$\begin{cases} dx = y\omega + \frac{x}{3t}dt, \\ dy = \frac{3x^2}{2}\omega + \frac{y}{2t}dt. \end{cases}$$

From this description we deduce the following relations between 2-forms

$$\begin{cases} dx \wedge dt = y\omega \wedge dt, \\ dy \wedge dt = \frac{3x^2}{2}\omega \wedge dt, \\ dx \wedge dy = -\frac{1}{2}\omega \wedge dt. \end{cases}$$

We can then further differentiate $\omega$ in this space obtaining

$$\begin{aligned} d\omega =& \frac{2}{3t}dx \wedge dy - \frac{2x}{3t^2}dt \wedge dy - \frac{1}{t}dy \wedge dx + \frac{y}{t^2}dt \wedge dx = \\ =& \left(-\frac{1}{3t} + \frac{x^3}{t^2} - \frac{1}{2t} - \frac{y^2}{t^2}\right)\omega \wedge dt = \\ =& \frac{1}{6}\omega \wedge dt \end{aligned}$$

The Gauss-Manin connection is the given by

$$\nabla(\omega) = \frac{1}{6t}\omega \otimes dt.$$

Analogously we can proceed to compute the differential of $x\omega$ in $\Omega_{\mathcal{E}}/\mathbb{C}^2$

$$d(x\omega) = dx \wedge \omega + xd\omega = \frac{2x}{3t}dx \wedge dy + \frac{1}{6t}x\omega \wedge dt = -\frac{1}{6t}x\omega \wedge dt.$$

Again the Gauss-Manin connection in this case is given by

$$\nabla(x\omega) = \left(-\frac{1}{6t}x\omega\right) \otimes dt.$$

With respect to the basis $\omega, x\omega$ we then have that the connection $\nabla$ is given by

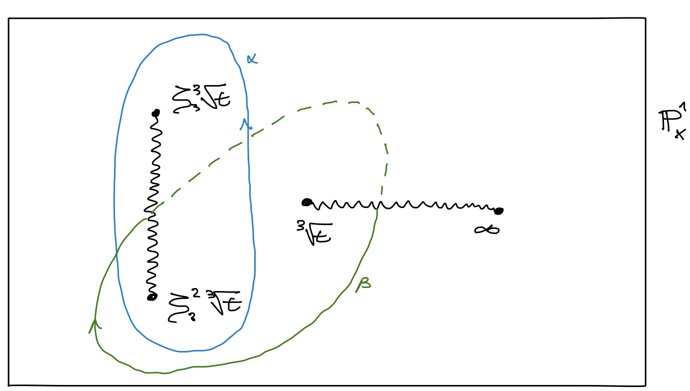
$$\nabla_t = \begin{pmatrix} 1/6t & 0 \\ 0 & -1/6t \end{pmatrix}.$$

Since we are working over $\mathbb{C}$, we can use the de Rham Theorem and shift our point of view on the connection from differential forms to homotopy theory. Recall that we have a perfect pairing given by

$$\begin{aligned} H^1_{dR}(\mathcal{E}_t, \mathbb{C}) \times H_1(\mathcal{E}_t, \mathbb{R}) &\longrightarrow \mathbb{C} \\ (\omega, \delta) &\longmapsto \int_\delta \omega \end{aligned}$$

The fibers of the relative de Rham sheaf $\mathcal{H}_{dR}(\mathcal{E}/S)$ are equal to the 2-dimensional vector space $H^1_{dR}(\mathcal{E}_t, \mathbb{C})$. The Gauss-Manin connection gives then a way to understand how to go from fiber to fiber. For a choice of $t$, we can picture the elliptic curve in the fiber of $t$ using the double-sheet visualization given by the 2:1 map on the $x$ coordinate

$$\begin{aligned} \mathcal{E}_t(\mathbb{C}) &\to \mathbb{P}^1(\mathbb{C}) \\ (x, y) &\mapsto x \end{aligned}$$



Consider a basis for homology given by the paths $\alpha, \beta$ in the picture where the solid arrows mean that the path is taken only over one sheet while the dotted ones pass to the other sheet. We then have

$$H_1(\mathcal{E}_t, \mathbb{Z}) = \mathbb{Z}\,\alpha + \mathbb{Z}\,\beta.$$

In order to see that $\alpha, \beta$ form a basis for the homology, we can use the intersection pairing and see that $\langle \alpha, \beta \rangle = 1$.

In order to understand how the 1-chains change from fiber to fiber, we can compute their period, namely

$$\int_\alpha \omega, \quad \int_\alpha x\omega, \quad \int_\beta \omega, \quad \int_\beta x\omega.$$

(1)

$$\int_\alpha \omega = 2\int_{\zeta_3^2 \sqrt[3]{t}}^{\zeta_3 \sqrt[3]{t}} \frac{dx}{\sqrt{x^3 - t}}$$

(2)

$$\int_\alpha \omega = 2\int_{\zeta_3^2 \sqrt[3]{t}}^{\zeta_3 \sqrt[3]{t}} \frac{xdx}{\sqrt{x^3 - t}}$$

(3)

$$\int_\beta \omega = 2\int_{\zeta_3^2 \sqrt[3]{t}}^{\sqrt[3]{t}} \frac{dx}{\sqrt{x^3 - t}}$$

(4)

$$\int_\beta \omega = 2\int_{\zeta_3^2 \sqrt[3]{t}}^{\sqrt[3]{t}} \frac{xdx}{\sqrt{x^3 - t}}$$

For the path $\alpha$ consider the change of variables

$$x(z) = \zeta_3^2 \sqrt[3]{t} + i\sqrt{3}\sqrt[3]{t} \cdot z$$

we then obtain

$$\begin{aligned}
\omega_\alpha &= \int_\alpha \omega = 2\frac{\sqrt{i}t^{-1/6}}{\sqrt[4]{3}} \int_0^1 \frac{dz}{\sqrt{z(z-1)(z+\zeta_3)}} \\
\eta_\alpha &= \int_\alpha x\omega = 2\frac{\sqrt{i}t^{1/6}}{\sqrt[4]{3}} \int_0^1 \frac{(\zeta_3^2 + i\sqrt{3}z)dz}{\sqrt{z(z-1)(z+\zeta_3)}} \\
\omega_\beta &= \int_\beta \omega = 2\frac{\sqrt{i}t^{-1/6}}{\sqrt[4]{3}} \int_0^{-\zeta_3} \frac{dz}{\sqrt{z(z-1)(z+\zeta_3)}} \\
\eta_\beta &= \int_\beta x\omega = 2\frac{\sqrt{i}t^{1/6}}{\sqrt[4]{3}} \int_0^{-\zeta_3} \frac{(\zeta_3^2 + i\sqrt{3}z)dz}{\sqrt{z(z-1)(z+\zeta_3)}}
\end{aligned}$$

From which we deduce

$$\frac{d}{dt}\begin{pmatrix} \omega_\alpha & \eta_\alpha \\ \omega_\beta & \eta_\beta \end{pmatrix} = \begin{pmatrix} \omega_\alpha & \eta_\alpha \\ \omega_\beta & \eta_\beta \end{pmatrix}\begin{pmatrix} -1/6 & 0 \\ 0 & 1/6 \end{pmatrix}.$$

2.2. **Legendre family.** Consider now the family of elliptic curves given by the Legendre family $\mathcal{E} : y^2 = x(x-1)(x-t)$ over $Spec(\mathbb{C}\left[t, \frac{1}{t(t-1)}\right])$ As before, we have a basis for the de Rham cohomology given by

$$\omega = \frac{dx}{y}, \quad \eta = \frac{xdx}{\eta}.$$

Let $P(x,t) = x(x-1)(x-t)$, then we have on $\Omega_{\mathcal{E}_t/\mathbb{C}}$

$$2ydy = P_x dx$$

where $P_x = 3x^2 - 2(t+1)x + t$. Using GP/Pari we can find $A, B$ rational functions in $t$ such that

$$AP + BP_x = 1.$$

In particular

$$\begin{cases} A = \left(\frac{-6t^2+6t-6}{t^4-2t^3+t^2}\right)x + \left(\frac{4t^3-3t^2-3t+4}{t^4-2t^3+t^2}\right), \\ B = \left(\frac{2t^2-2t+2}{t^4-2t^3+t^2}\right)x^2 + \left(\frac{-2t^3+t^2+t-2}{t^4-2t^3+t^2}\right)x + \frac{1}{t}. \end{cases}$$

We can then take

$$\omega = Aydx + 2Bdy$$

in this way we have the usual relations

$$\begin{cases} dx = y\omega, \\ dy = \frac{1}{2}P_x\omega. \end{cases}$$

In the full module $\Omega^1_{\mathcal{E}/\mathbb{C}}$ we have

$$2ydy = P_xdx + P_tdt$$

with $P_t = -x(x-1)$. From this we obtain

$$\begin{cases} dx = y\omega - BP_tdt, \\ dy = \frac{1}{2}P_x\omega + \frac{1}{2}AP_tydt. \end{cases}$$

From this description we deduce the following relations between 2-forms

$$\begin{cases} dx \wedge dt = y\omega \wedge dt, \\ dy \wedge dt = \frac{1}{2}P_x\omega \wedge dt, \\ dx \wedge dy = \frac{1}{2}P_t\omega \wedge dt. \end{cases}$$

We can then further differentiate $\omega$ in this space obtaining

$$\begin{aligned} d\omega =& Ady \wedge dx + A_tydt \wedge dx + 2B_xdx \wedge dy + 2B_tdt \wedge dy = \\ =& \left(-\frac{1}{2}AP_t - A_tP + B_xP_t - B_tP_x\right)\omega \wedge dt = \\ =& \left[\left(\frac{-5t^2+5t-5}{t^4-2t^3+t^2}\right)x^3 + \left(\frac{8t^3+3t^2-9t+14}{2t^4-4t^3+2t^2}\right)x^2 + \left(\frac{-8t^3+9t^2-3t-4}{2t^4-4t^3+2t^2}\right)x + \frac{1}{t}\right]\omega \wedge dt \end{aligned}$$

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In order to obtain a representative for a de Rham class in the usual basis, we want to rewrite $x^3\omega$ and $x^2\omega$ as an exact form plus a linear combination of $\omega$ and $\omega$. We can use the fact that every function on $\mathcal{E}_t$ can be written as $C + Dy$ where $C, D$ are polynomials in $x$. Then to have an exact form we need to have

$$(C + Dy)\omega = d(E + Fy) = E'dx + F'ydx + Fdy = (E'y + F'P + \frac{1}{2}FP')\omega$$

and then

$$\begin{cases} C = \frac{1}{2}FP' + F'P, \\ D = E'. \end{cases}$$

Solving for $x^3$ and $x^2$ we obtain

$$\begin{aligned} x^3\omega =& d\left(\frac{2}{5}xy + \frac{8}{15}(1+\lambda)y\right) + \left(\frac{8}{15}t^2 + \frac{7}{15}t + \frac{8}{15}\right)x\omega - \left(\frac{4}{15}t^2 + \frac{4}{15}t\right)\omega, \\ x^2\omega =& d\left(\frac{2}{3}y\right) + \frac{2}{3}(t+1)x\omega - \frac{1}{3}t\omega. \end{aligned}$$

We can finally conclude that the Gauss-Manin connection is given by

$$\nabla(\omega) = -\frac{1}{2t(t-1)}x\omega + \frac{1}{2(t-1)}.$$

To calculate $\nabla(x\omega)$ we can use the differential rule

$d(x\omega) = xd\omega + dx \wedge \omega =$

$$= \left[ \left( \frac{-7t^2+7t-7}{t^4-2t^3+t^2} \right) x^4 + \left( \frac{12t^3+5t^2-15t+22}{2t^4-4t^3+2t^2} \right) x^3 + \left( \frac{-14t^3+15t^2-3t-8}{2t^4-4t^3+2t^2} \right) x^2 + \frac{2}{t} x \right] \omega \wedge t.$$

Again we can calculate the de Rham class of $x^4\omega$

$$x^4\omega = d\left(x^2 + \left(\frac{12}{35}t + \frac{12}{35}\right)x + \left(\frac{16}{35}t^2 + \frac{46}{105}t + \frac{16}{35}\right)\right) - \left(\frac{16}{35}t^3 + \frac{8}{21}t^2 + \frac{8}{21}t + \frac{16}{35}\right)x\omega - \left(\frac{8}{35}t^3 + \frac{23}{105}t^2 + \frac{8}{35}t\right)\omega.$$

We then conclude

$$\nabla(x\omega) = \frac{-3t+2}{2t(t-1)}x\omega + \frac{1}{2(t-1)}\omega.$$

The matrix associated to the Gauss-Manin connection with respect to the basis $\omega, x\omega$ is

$$\nabla \cdot \begin{pmatrix} \omega \\ x\omega \end{pmatrix} = \frac{1}{2(t-1)} \begin{pmatrix} 1 & -\frac{1}{t} \\ 1 & \frac{-3t+2}{t} \end{pmatrix} \begin{pmatrix} \omega \\ x\omega \end{pmatrix}$$

2.3. **The third family.** Consider now this third family of elliptic curves given by $\mathcal{E} : y^2 = (x^2 - t)(x-1)$ over $Spec(\mathbb{C}\left[t, \frac{1}{t}\right])$ As before, we have a basis for the de Rham cohomology given by

$$\omega = \frac{dx}{y}, \quad \eta = \frac{xdx}{\eta}.$$

Let $P(x,t) = (x^2-t)(x-1)$, then we have on $\Omega_{\mathcal{E}_t/\mathbb{C}}$

$$2ydy = P_x dx$$

where $P_x = 3x^2 - 2x - t$. Using GP/Pari we can find $A, B$ rational functions in $t$ such that

$$AP + BP_x = 1.$$

In particular

$$\begin{cases} A = \left(\frac{-9t-3}{2t^3-4t^2+2t}\right)x + \left(\frac{-3t+1}{t^3-2t^2+t}\right), \\ B = \left(\frac{3t+1}{2t^3-4t^2+2t}\right)x^2 + \frac{1}{2t^2-2t}x + \left(\frac{-t-1}{t^2-2t+1}\right). \end{cases}$$

We can then take

$$\omega = Aydx + 2Bdy$$

in this way we have the usual relations

$$\begin{cases} dx = y\omega, \\ dy = \frac{1}{2}P_x\omega. \end{cases}$$

In the full module $\Omega^1_{\mathcal{E}/\mathbb{C}}$ we have

$$2ydy = P_x dx + P_t dt$$

with $P_t = -x + 1$. From this we obtain

$$\begin{cases} dx = y\omega - BP_t dt, \\ dy = \frac{1}{2}P_x\omega + \frac{1}{2}AP_t ydt. \end{cases}$$

From this description we deduce the following relations between 2-forms

$$\begin{cases} dx \wedge dt = y\omega \wedge dt, \\ dy \wedge dt = \frac{1}{2}P_x\omega \wedge dt, \\ dx \wedge dy = \frac{1}{2}P_t\omega \wedge dt. \end{cases}$$

We can then further differentiate $\omega$ in this space obtaining

$$
\begin{aligned}
d\omega =& Ady \wedge dx + A_t y dt \wedge dx + 2B_x dx \wedge dy + 2B_t dt \wedge dy = \\
&= \left(-\frac{1}{2}AP_t - A_t P + B_x P_t - B_t P_x\right)\omega \wedge dt = \\
&= \left[\left(\frac{-9t-3}{4t^3-8t^2+4t}\right)x^2 + \left(\frac{5t+3}{4t^3-8t^2+4t}\right)x + \frac{t}{t^2-2t+1}\right]\omega \wedge dt,
\end{aligned}
$$

$$
\begin{aligned}
d(x\omega) &= xd\omega + dx \wedge \omega = \\
&= \left[\left(\frac{-15t-5}{4t^3-8t^2+4t}\right)x^3 + \left(\frac{9t+7}{4t^3-8t^2+4t}\right)x^2 + \left(\frac{4t^2+3t-1}{2t^3-4t^2+2t}\right)x + \left(\frac{-t-1}{t^2-2t+1}\right)\right]\omega \wedge t.
\end{aligned}
$$

In order to obtain a representative for a de Rham class in the usual basis, we want to rewrite $x^3\omega$ and $x^2\omega$ as an exact form plus a linear combination of $\omega$ and $\omega$. We can use the fact that every function on $\mathcal{E}_t$ can be written as $C + Dy$ where $C, D$ are polynomials in $x$. Then to have an exact form we need to have

$$(C + Dy)\omega = d(E + Fy) = E'dx + F'ydx + Fdy = (E'y + F'P + \frac{1}{2}FP')\omega$$

and then

$$\begin{cases} C = \frac{1}{2}FP' + F'P, \\ D = E'. \end{cases}$$

Solving for $x^3$ and $x^2$ we obtain

$$
\begin{aligned}
x^3\omega =& d\left(\frac{2}{5}xy + \frac{8}{15}y\right) + \left(\frac{3}{5}t + \frac{8}{15}\right)x - \frac{2}{15}t\omega, \\
x^2\omega =& d\left(\frac{2}{3}y\right) + \frac{2}{3}x\omega + \frac{1}{3}t\omega.
\end{aligned}
$$

We can finally conclude that the Gauss-Manin connection is given by

$$
\begin{aligned}
\nabla(\omega) &= \frac{-1}{4t^2-4t}x\omega + \frac{1}{4t-4}\omega, \\
\nabla(x\omega) &= \frac{-1}{4t-4}x\omega + \frac{1}{4t-4}\omega.
\end{aligned}
$$

The matrix associated to the Gauss-Manin connection with respect to the basis $\omega, x\omega$ is

$$\nabla \cdot \begin{pmatrix} \omega \\ x\omega \end{pmatrix} = \frac{1}{4(t-1)} \begin{pmatrix} 1 & -\frac{1}{t} \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \omega \\ x\omega \end{pmatrix}$$

## 3. Hypergeometric series

The hypergeometric series is defined as:

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n n!} z^n,$$

where $(a)_n$ is the Pochhammer symbol (or rising factorial) defined by:

$$(a)_n = a(a+1)(a+2)\dots(a+n-1), \quad \text{with} \quad (a)_0 = 1.$$

This series converges for $|z| < 1$ and can be analytically continued to other values of $z$, except for certain singularities.

The integral representation of the hypergeometric function ${}_2F_1(a, b; c; z)$ is given by:

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-b-1}(1-tz)^{-a}\, dt,$$

where the parameters satisfy $\Re(c) > \Re(b) > 0$ and $|z| < 1$.

In particular we can then express the value of periods of Elliptic Curves in terms of these hypergeometric functions. First of all consider the Legendre family $y^2 = x(x-1)(x-\lambda)$ we then have

$$\begin{aligned}\int_0^1 \frac{dx}{\sqrt{x(x-1)(x-\lambda)}} &= \frac{\pi}{\sqrt{\lambda}} \cdot {}_2F_1\left(\frac{1}{2},\frac{1}{2},1;\frac{1}{\lambda}\right) = \Pi_1(\lambda)\\ \int_0^\lambda \frac{dx}{\sqrt{x(x-1)(x-\lambda)}} &= \pi \cdot {}_2F_1\left(\frac{1}{2},\frac{1}{2},1;\lambda\right) = \Pi_2(\lambda)\\ \int_0^1 \frac{xdx}{\sqrt{x(x-1)(x-\lambda)}} &= \frac{\pi}{2\sqrt{\lambda}} \cdot {}_2F_1\left(\frac{3}{2},\frac{1}{2},2;\frac{1}{\lambda}\right) = \Pi_3(\lambda)\\ \int_0^\lambda \frac{xdx}{\sqrt{x(x-1)(x-\lambda)}} &= \frac{\lambda\pi}{2} \cdot {}_2F_1\left(\frac{3}{2},\frac{1}{2},2;\lambda\right) = \Pi_4(\lambda).\end{aligned}$$

We then define the period matrix

$$\Pi_\lambda = \begin{pmatrix}\Pi_1(\lambda) & \Pi_2(\lambda)\\ \Pi_3(\lambda) & \Pi_4(\lambda)\end{pmatrix}$$

Consider now a general family $\mathcal{E} : y^2 = (x-e_1(t))(x-e_2(t))(x-e_3(t))$. If we want to compute the periods going from $e_1$ to $e_2$ and from $e_1$ to $e_3$ we can perform the change of variables

$$x = (e_2 - e_1)z + e_1$$

obtaining

$$\begin{aligned}\int_{e_1}^{e_2} \frac{dx}{\sqrt{(x-e_1)(x-e_2)(x-e_3)}} &= \frac{1}{\sqrt{e_2-e_1}}\Pi_1(\lambda_e),\\ \int_{e_1}^{e_3} \frac{dx}{\sqrt{(x-e_1)(x-e_2)(x-e_3)}} &= \frac{1}{\sqrt{e_2-e_1}}\Pi_2(\lambda_e),\\ \int_{e_1}^{e_2} \frac{xdx}{\sqrt{(x-e_1)(x-e_2)(x-e_3)}} &= \sqrt{e_2-e_1}\cdot\Pi_3(\lambda_e) + \frac{e_1}{\sqrt{e_2-e_1}}\Pi_1(\lambda_e),\\ \int_{e_1}^{e_3} \frac{xdx}{\sqrt{(x-e_1)(x-e_2)(x-e_3)}} &= \sqrt{e_2-e_1}\cdot\Pi_4(\lambda_e) + \frac{e_1}{\sqrt{e_2-e_1}}\Pi_2(\lambda_e),\end{aligned}$$

with $\lambda_e = \frac{e_3-e_1}{e_2-e_1}$. The period matrix is then given by

$$\Pi(\mathcal{E}) = \begin{pmatrix}\frac{1}{\sqrt{e_2-e_1}} & 0\\ \frac{e_1}{\sqrt{e_2-e_1}} & \sqrt{e_2-e_1}\end{pmatrix}\Pi_{\lambda_e}$$

3.1. **Asymptotic behavior for $\lambda \to 0$.** The hypergeometric function ${}_2F_1(a,b,c;z)$ is a well studied function and in particular we have access to its asymptotic behavior for small values of $z$.

$$\begin{aligned}{}_2F_1(a,b,c;z) &= 1 + \frac{ab}{c}z + O(z^2),\\ {}_2F_1(a,b,c;\frac{1}{z}) &= \end{aligned}$$

## References

[Sil09] Joseph H. Silverman. *The Arithmetic of Elliptic Curves*, volume 106 of *Graduate Texts in Mathematics*. Springer, New York, 2 edition, 2009. First edition published in 1986.